



Day-3

Ram & Storage Theory

Types of Ram (DDR3,DDR4,DDR5) and storage (HDD vs. SSD NVMe .Remove and reinstall Ram and storage devices.

Q-1 :- Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Ans – Comparison of DDR3,DDR4,DDR5 is as below.

DDR3 vs DDR4 vs DDR5 RAM – Comparison

Feature	DDR3	DDR4	DDR5
Speed	800 – 2133 MT/s (million transfers per second)	1600 – 3200 MT/s (million transfers per second)	4800 – 8800+ MT/s (million transfers per second)
Voltage	1.5 V (typical)	1.2 V	1.1 V
Performance	Slowest	Faster than DDR3	Fastest of the three
Power Use	Higher	Lower	Lowest voltage; better power efficiency
Capacity	Usually smaller (up to 16 GB per stick common)	Larger than DDR3 (up to 32 GB per stick common)	Supports higher capacities (up to 64 GB+ per stick common)
Typical Use Cases	Older desktop PCs and laptops	Many older and mid-range PCs	Newer desktops, laptops, and gaming PCs
Main Advantage	Affordable for older systems	Good balance of speed and cost	High speed and better multitasking



Simple Explanation-
DDR3: An older type of RAM used in older computers.
DDR4: Faster and more power-efficient than DDR3, commonly found in many PCs.
DDR5: The newest generation of the three, offering higher speeds and improved efficiency for modern computing and gaming.

Important : DDR3, DDR4, and DDR5 are not interchangeable. Your motherboard must support the specific RAM generation you want to use.

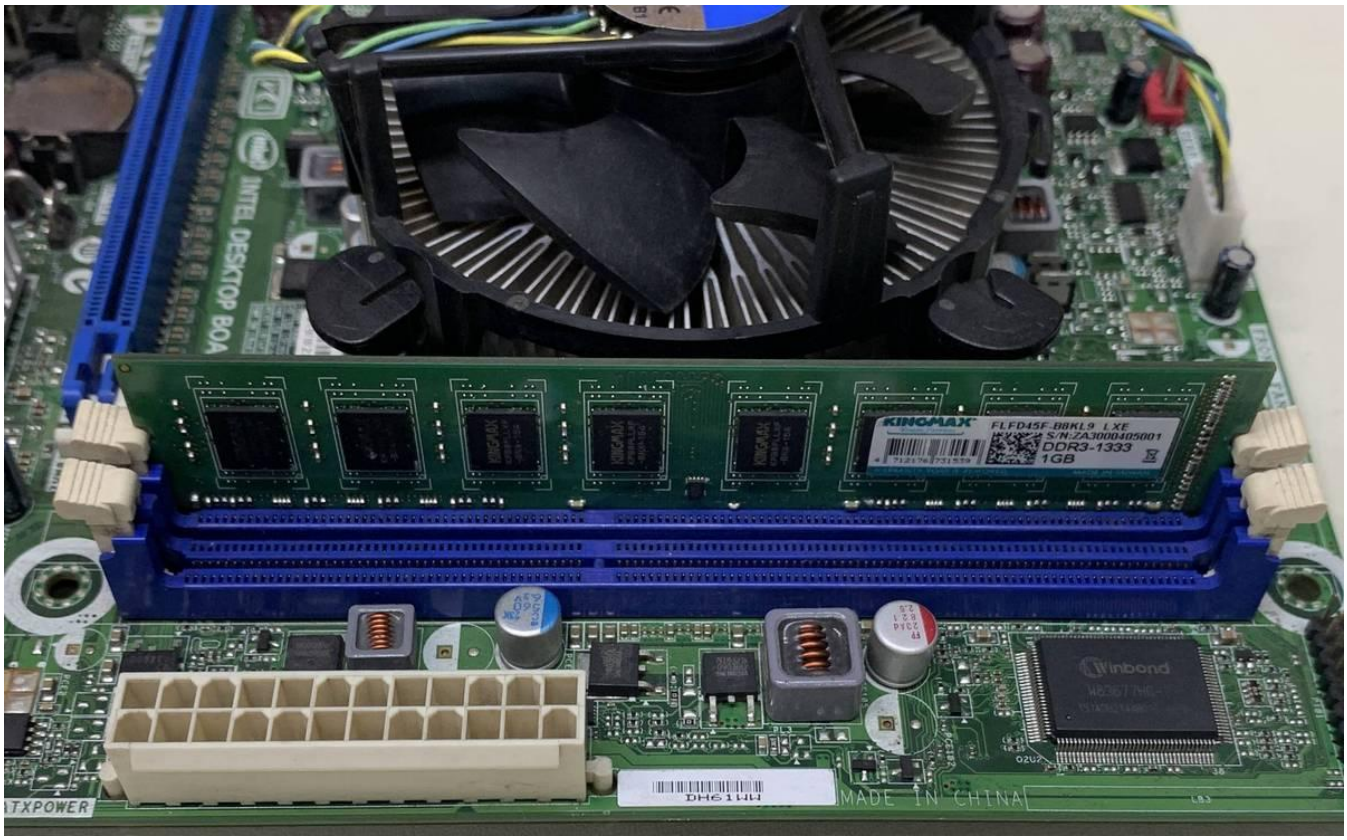
Note : RAM speeds are measured in MT/s (million transfers per second). Actual speeds depend on the RAM model and the motherboard.

Q-2 :-Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.

Ans-Here is a simple step-by-step guide for the practical computer hardware task.

1. Locate and photograph the RAM



2. Steps to remove and reinstall RAM

-1. Turn off the computer. Unplug the power cable and disconnect all accessories.
-2. Open the case or laptop cover. Follow your device manual.
-3. Find the RAM slots. RAM is a long, thin circuit board attached to the motherboard.
-4. Take a clear photo. Photograph the installed RAM before removing it.
-5. Remove the RAM carefully. Release the clips if your device has them. Avoid – touching the gold contacts.

-6. Reinstall the RAM. Align the notch correctly and press gently until it is fully seated.
7. Close the cover and reconnect power. Turn on the computer and check that it boots successfully.

Note: Never remove or install RAM while the computer is connected to power. Some laptops have built-in batteries that require additional precautions; follow the manufacturer's instructions.

Verification: The computer started successfully after reinstallation. (Write this only if your computer booted successfully.)

Conclusion: RAM was reinstalled correctly, and the system was checked for proper startup.

Q 3 :-List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Ans-HDD vs SSD vs NVMe is as below.

HDD vs SSD vs NVMe – Key Differences & Best Use Cases

Feature	 HDD (Hard Disk Drive)	 SSD (SATA SSD)	 NVMe (NVMe SSD)
1. Speed	**Slow** (around 80–160 MB/s)	**Fast** (around 500 MB/s – 550 MB/s)	**Very fast** (1,500 MB/s – 7,000 MB/s+)
2. Durability	**Less durable** (has moving parts – spinning disks)	**More durable** (no moving parts)	**More durable** (no moving parts)
3. Cost & Capacity	**Cheapest** and offers the largest storage space (e.g., 1TB, 2TB, 4TB+)	**More expensive than HDD** (e.g., 256GB – 2TB)	**Usually more expensive than SSD, but offers the highest speed** (e.g., 512GB – 4TB+)
4. Best Use Case	Storing large files like photos, videos, and backups	Opening apps and loading your system quickly	Gaming, video editing, and loading large files/games very quickly
5. Example App / Use Case	**File storage & backup** (e.g., keeping family photos and videos)	**Google Chrome** (faster startup and quick browsing)	**FIFA / modern PC games** (quick loading and smooth gameplay)

HDD – Simple Take

- Slower, less durable, but cheapest.
- Best for storing lots of data at a low cost.
- Great for photos, videos, and backups.

SSD – Simple Take

- Faster, more durable than HDD.
- Good balance of speed and price.
- Great for everyday apps and faster boot times.

NVMe – Simple Take

- Much faster than SSD.
- Durable and high performance.
- Best for gaming, video editing, and large file transfers.

Note: NVMe is a type of SSD technology that uses a faster connection (PCIe) than traditional SATA SSDs.

Simple Explanation :-

HDD : Best for storing large amounts of data at a low cost.
-Stores photos and videos cheaply.

SSD: Best for everyday tasks, such as starting your computer and opening applications quickly.
-Opens apps faster, like Chrome.

NVMe SSD: Best for tasks that need very high speed, such as loading games and transferring large files.
-Loads games very quickly, like FIFA.

Note: NVMe is a type of SSD technology that uses a faster connection than traditional SATA SSDs.

Q-4 :-Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.

Ans-Remove and Reinstall a Storage Drive.



Simple steps

1. Turn off the computer and disconnect the power cable.
2. Open the computer case or laptop cover using the device manual.
3. Find the storage drive. It may be an HDD, SATA SSD, or M.2 SSD.
4. Take a clear photo of the drive and its connector.
5. Remove the drive carefully by disconnecting its cable or removing its holding screw, depending on the type.
6. Reinstall the drive securely in its original position.
7. Close the cover and turn on the computer.

8. Check the storage. Make sure the drive appears in File Explorer or your computer's BIOS/UEFI.

Note: Never remove a drive while the computer is powered on. If you do not have permission or experience opening the device, do not open it.